Visualization Tools for Structural Analysis

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CS 01

INTRODUCTION & PROBLEM

- KPFF is a structural engineering firm with more than 64 years of experience.
- KPFF engineers use **OpenSees**, a **command-line based** seismic simulation tool for structural analysis.
- This **lack of visualization** slows workflows.

EXISTING SOLUTIONS

- Tools like **RISA-3D®** and **ETABS®** focus on usability, but lack advanced analysis.
- **Perform3D®** is powerful but outdated and unintuitive.
- OpenSees has no GUI, requiring manual text output review and third-party software.

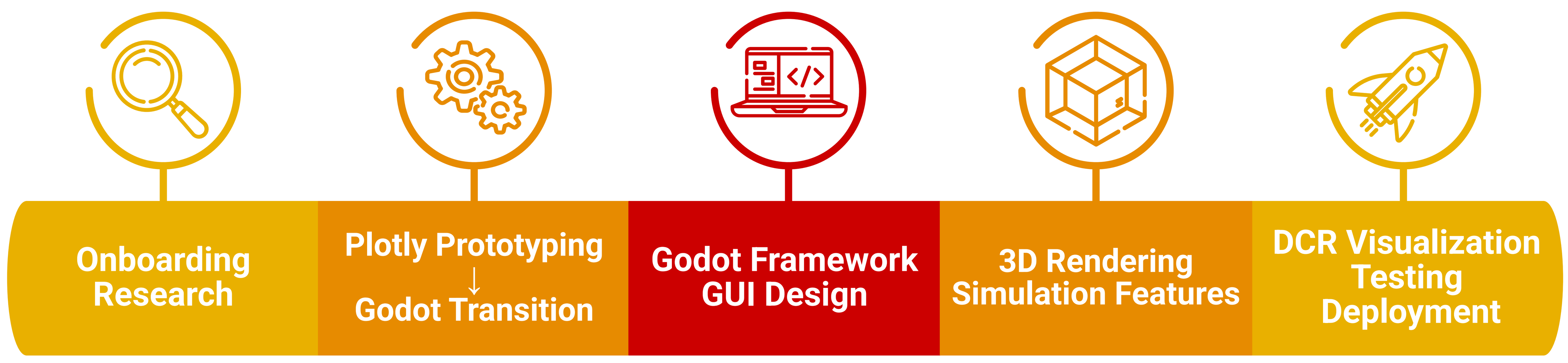
PROJECT GOAL

- Deliver an interactive 3D visualization tool that bridges the gap between OpenSees' power and intuitive usability – **all within a single interface**.
- This will enable engineers to visually review models and outputs before and after OpenSees simulations.

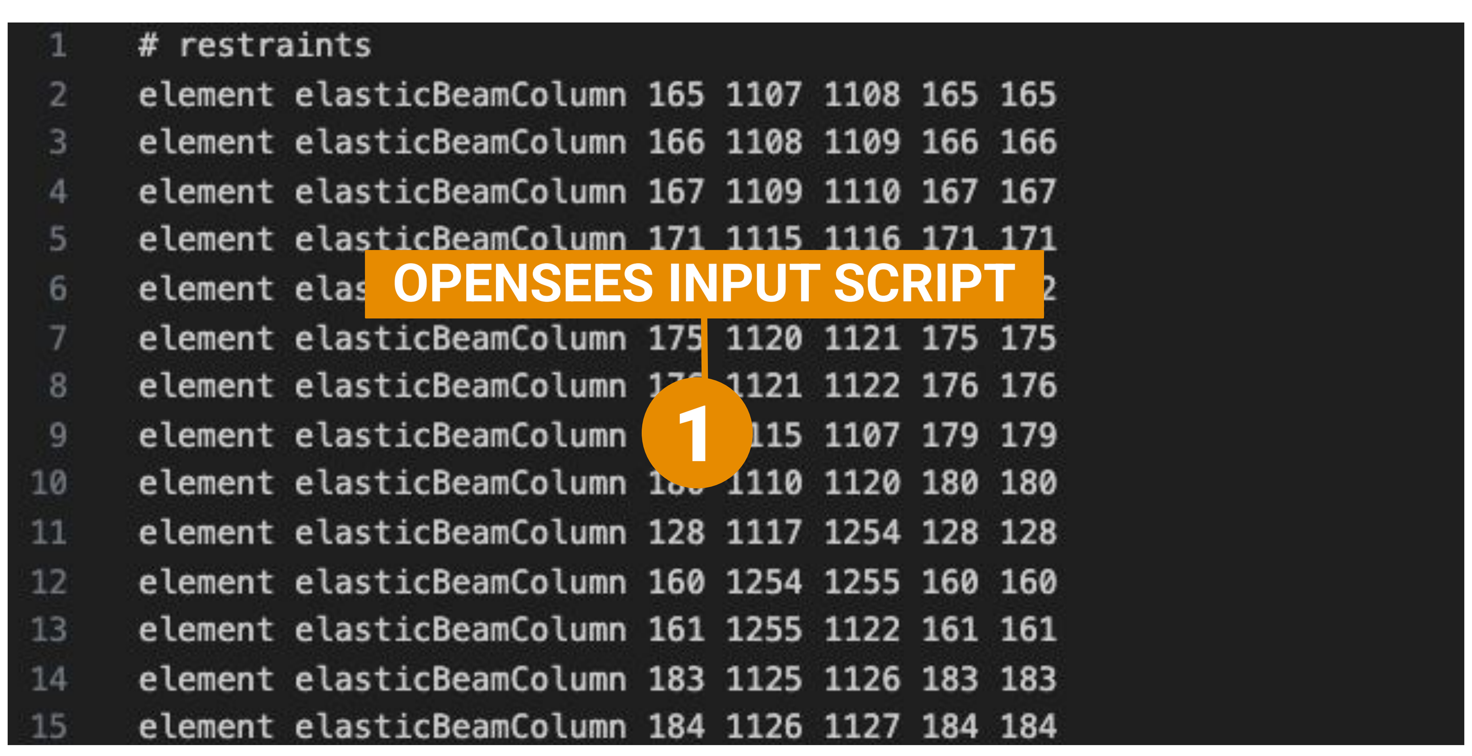
TECH STACK

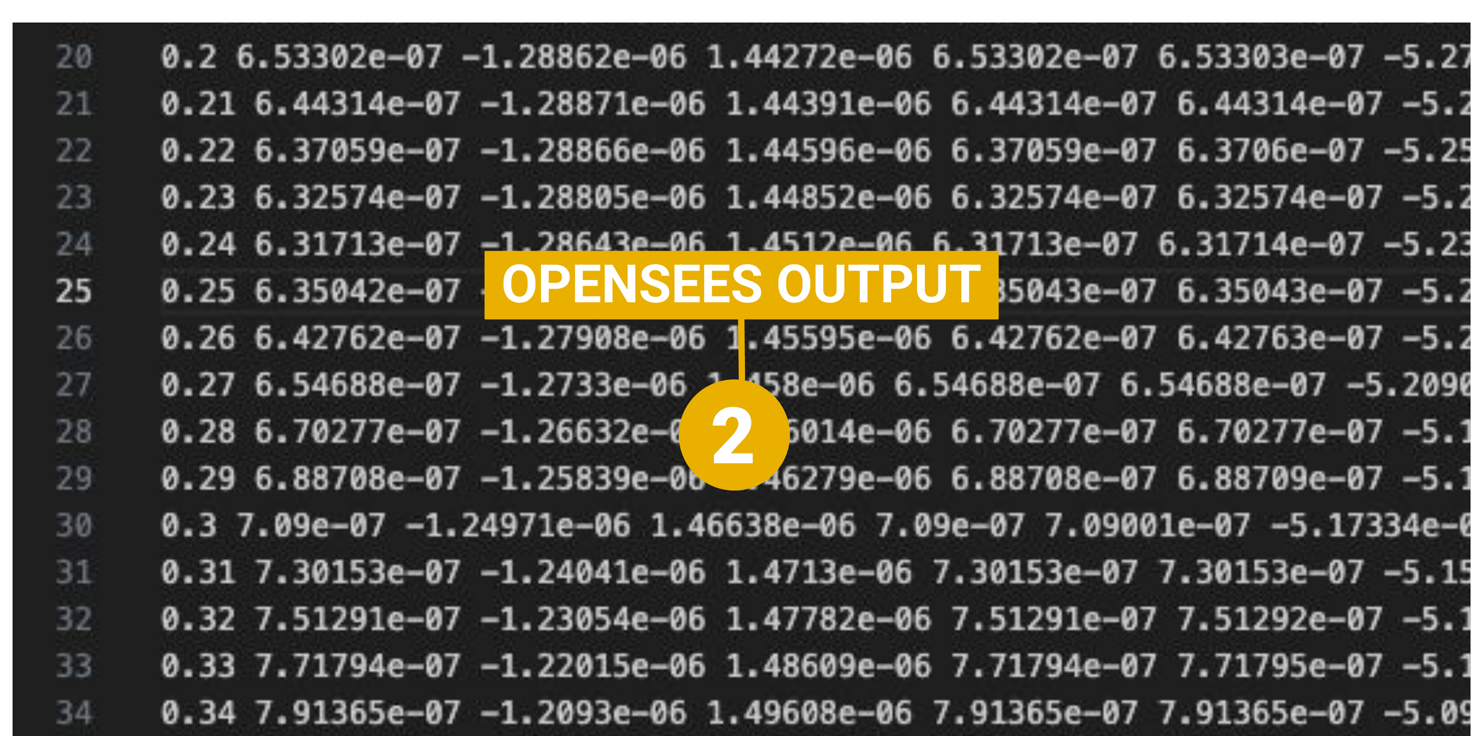
GODOT
Game engine

DEVELOPMENT ROADMAP



DESIGN IMPLEMENTATION AND FEATURES





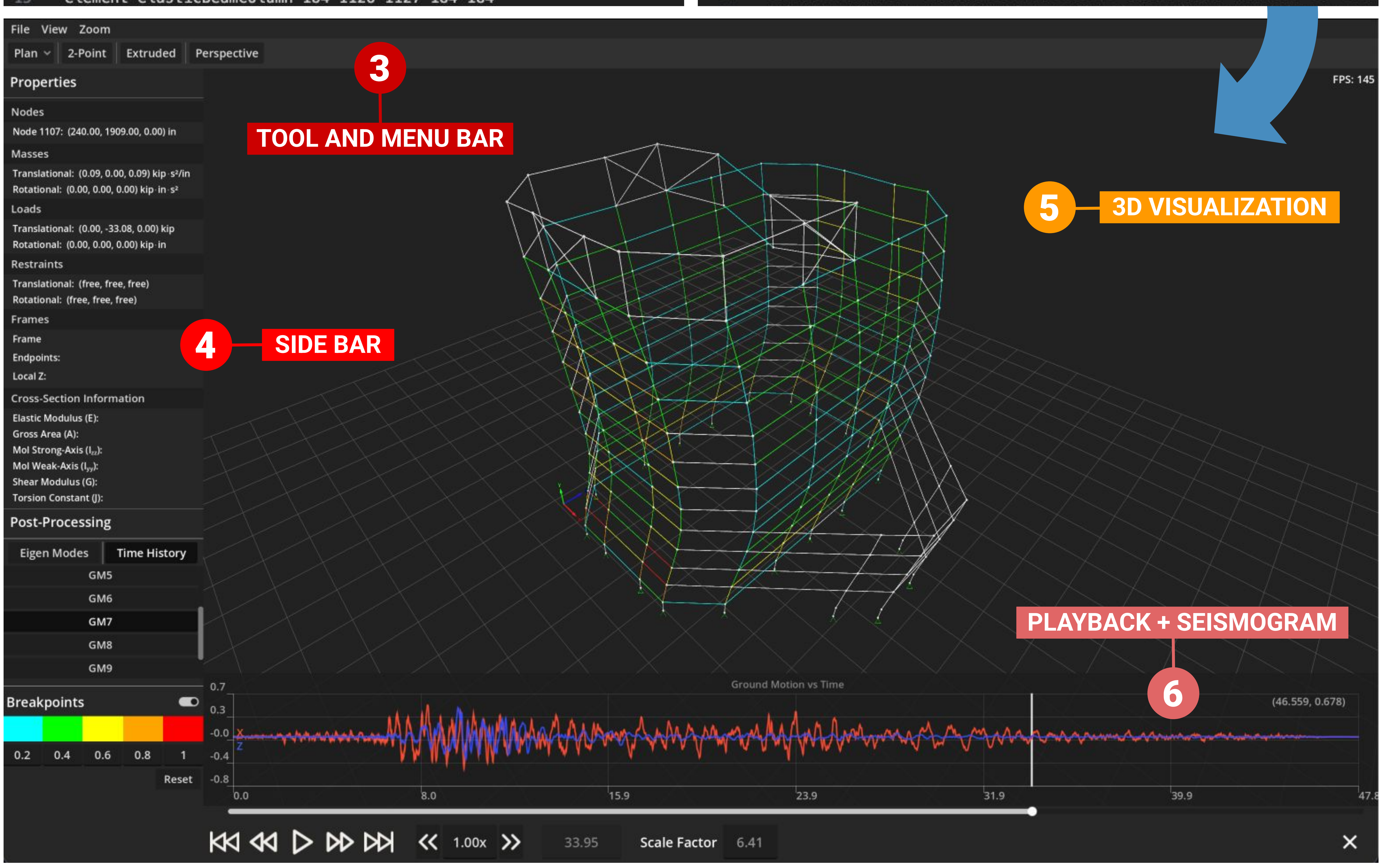


FIGURE KEY

Supplied by KPFF:

1. Script defining model properties.

2. Raw analysis output data including node displacements, forces and response values.

Built by our team:

3. Controls for **file import**, camera **views** and settings, and adjustable **zoom** settings.

4. Togglable **properties** for elements, masses, loads, restraints, and cross-section. Real-time **DCR** color scaling and adjustable **breakpoints**. **Eigen modes** and **time history** analyses.

5. Interactive 3D model viewport with real-time **pan**, **rotate**, **zoom** and **inspection** for pre- and post-processing.

6. Playback controls (**time-step**, **speed**, **progress**) with adjustable x/z Ground Motion vs Time plot.

FUTURE WORK

- Add **group** and **elevation** view modes.
- **OpenSees backend integration.**
- 3D cityscape overlay for client context.
- Display **node** and **beam tags** on the model.
- Show **directional load arrows** for clarity.
- Add plastic hinges, dampers, and nonlinear wall elements, as well as build tools for pushover analysis and **performance checking.**

REFERENCES & ACKNOWLEDGEMENTS

- Special thanks to KPFF engineers and UCI ICS faculty for their guidance, technical feedback, and ongoing collaboration.
- We gratefully acknowledge the open-source software tools used in our development: OpenSees, Godot Engine, LD2Studio - graph2D.

QUESTIONS? CONTACT US!